

Good morning/ afternoon/ evening to everyone present here.

I am _____ and this is my team. As highlighted in our poster, mathematics is the backbone that allows us to design efficient, secure, and accurate computational solutions.

It all starts with **logic** and **set theory**, forming the foundation of every algorithm through concepts of “if-then,” “and,” “or,” and “not.” Set theory helps us organise and manage data, enabling key operations. **Linear algebra** builds on this, giving us tools to work with **vectors** and **matrices**—essential for machine learning and computer graphics.

Calculus is at the heart of optimisation, driving machine learning models to minimise errors using methods like gradient descent. This is how computers ‘learn’ and improve their predictions. **Probability** and statistics help algorithms handle uncertainty, making sense of noisy or incomplete data. **Bayesian statistics**, for example, are used in spam filters to determine the chance that a message is unwanted.

Graph theory models relationships, helping algorithms understand connections in networks such as social media or logistics. **Number theory** is critical for cryptography, with algorithms like RSA relying on its principles to keep our communications secure.

Introducing **randomness** can make some algorithms, like quicksort, more efficient and reliable. Lastly, **machine learning** leverages probability to make predictions—such as forecasting a 95% chance of rain.

In summary, **mathematics** is not just a set of tools, but a rigorous framework powering every aspect of algorithms. By mastering these mathematical foundations, we can innovate and solve real-world problems as computer scientists.

Thank you for your attention, and I hope this presentation inspires you to appreciate **mathematics role** in **Algorithms**.